

PS-036 · PQC DNSSEC MTL Benchmark

Using a Reference Implementation

Team RootOps
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Problem Description

The challenge

DNSSEC protects DNS responses with digital signatures. Future quantum-capable attackers could threaten widely used classical public-key cryptography.

Why it matters

Post-quantum signatures can introduce larger signatures and additional signing or verification costs, potentially affecting DNS response size and latency.

Project objective

- Benchmark PQC-based DNSSEC using a reference implementation.
- Compare ML-DSA-44 signing with an MTL-based approach using ML-DSA-44.
- Measure cryptographic cost, signature size, and end-to-end DNS resolution performance.

Background

DNSSEC

Adds digital signatures to DNS data so resolvers can validate authenticity and integrity through a chain of trust.

Post-Quantum Cryptography

Cryptographic algorithms designed to withstand attacks from quantum computers. ML-DSA-44 is the signature scheme used in our comparison.

Merkle Tree Ladder

An MTL-based approach can amortize expensive public-key signatures across multiple records using hash-based authentication structures. The benchmark measures its trade-offs.

Literature & RFC Review

Relevant RFCs and Standards

- RFC 4033: Introduces DNSSEC and its security requirements, including data authentication and integrity.
- RFC 4034: Defines DNSSEC resource records such as DNSKEY, DS, RRSIG, and NSEC.
- RFC 4035: Specifies DNSSEC protocol modifications and validation procedures.
- RFC 9904: Describes the process for maintaining DNSSEC cryptographic algorithm recommendations.
- NIST FIPS 204: Standardizes ML-DSA, a digital signature algorithm designed to resist quantum-computer attacks.

Research Gap: These standards establish the foundations for DNSSEC and post-quantum signatures, while our project benchmarks ML-DSA-44 against an MTL-based approach to evaluate signing time, verification time, signature size, and DNS resolution performance.

Proposed Solution

We propose a benchmark testbed using a reference implementation to evaluate Post-Quantum Cryptography (PQC) in DNSSEC.

Key Components

- DNS Hierarchy: Simulates root, TLD, and authoritative DNS servers.
- Baseline Approach: Uses ML-DSA-44 signatures for DNSSEC records.
- MTL Approach: Uses a Merkle Tree Ladder (MTL) to reduce repeated public-key signing overhead.
- Validation: Tests DNS resolution and verifies the DNSSEC/PQC chain.
- Benchmarking: Measures signing time, verification time, signature size, and DNS resolution performance.

Objective: Compare both approaches under controlled conditions to understand their performance and practical trade-offs.

Watch our **team introduction** video: here

Optimization Proposed

The proposed optimization explores whether an MTL-based approach can improve the efficiency of PQC-based DNSSEC.

Parameter	Baseline: ML-DSA-44	MTL-Based Approach
Signing time	Cost to generate DNSSEC signatures	Compare signing overhead
Verification time	Cost to validate signatures / chain	Assess resolver-side cost
Signature size	Bytes added to signed DNS data	Estimate response-size impact
DNS resolution	Success, latency, and failure rate	Check end-to-end practicality

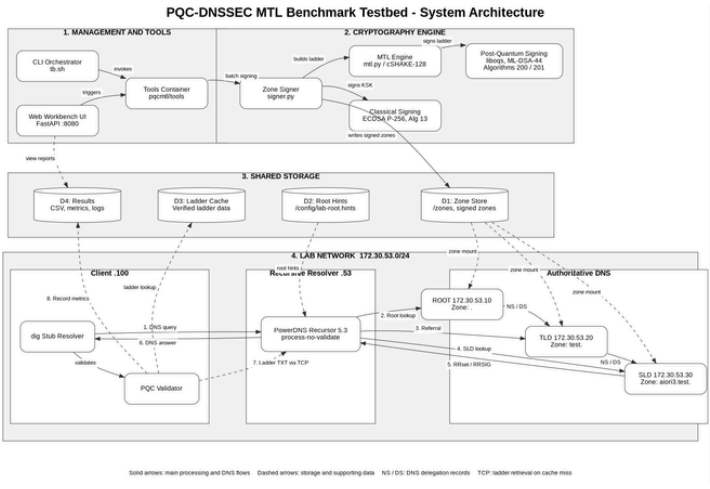
Evaluation Strategy

- Test small, medium, and large DNS datasets.
- Measure signing and verification time.
- Compare signature sizes and DNS resolution latency.
- Validate correctness and report any trade-offs.

Expected Benefit: Reduced repeated signing overhead and smaller authentication data, subject to experimental validation.

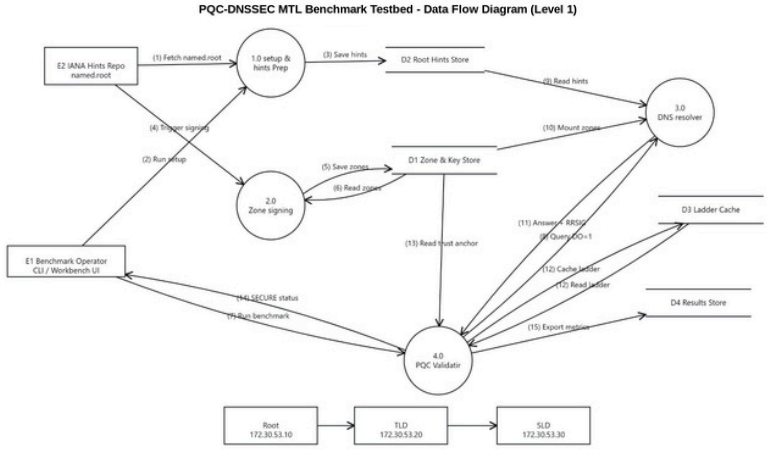
Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.



Deployment Timeline

Phase 1 — Environment Setup

- Configure the reference implementation and required dependencies.
- Set up the simulated DNS hierarchy.

Phase 2 — Implementation

- Generate DNSSEC-signed zones using ML-DSA-44 and MTL modes.
- Configure the DNS resolver and authoritative servers.

Phase 3 — Validation

- Test DNS query resolution.
- Verify the DNSSEC/PQC chain and confirm successful responses.

Phase 4 — Benchmarking

- Execute workloads using small, medium, and large datasets.
- Record signing time, verification time, signature size, and DNS resolution performance.

Phase 5 — Analysis and Reporting

- Compare the two approaches.
- Identify performance bottlenecks and trade-offs.
- Document the findings and prepare the final report.

Milestone: A functioning testbed with reproducible benchmark results and a comparative analysis.

Bibliography I

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Thank You for Your Attention!